UNIVERSITY OF APPLIED SCIENCES
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Installation Instructions and User Guide

Project: MAUSI

Phase: Release

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1 Introduction

1.1 Purpose

This document describes how to install (build and flash) the firmware of the MAUSI Micromouse system and how to operate the robots, including the “Pac-Man” game mode. It is intended for developers and operators (students) who set up and test the system, as well as for end users who interact with it during demonstrations. The MAUSI system consists of several Micromice based on the Pololu 3pi+ 2040 platform, each extended with an ESP32 module for wireless communication and game-controller input.

1.2 Definitions, Acronyms and Abbreviations

Term	Definition
MAUSI	Project name of the multi-agent Micromouse system
RP2040	Main microcontroller of the Pololu 3pi+ 2040 robot
ESP32	Wireless co-processor
UART	Serial interface between the RP2040 and the ESP32
ToF	Time-of-Flight distance sensor (wall detection)
UF2	Bootloader file format used to flash the RP2040
BOOTSEL	Bootloader-select mode/button of the RP2040
OLED	On-board 128x64 pixel display
HUD	Head-Up Display (game overlay shown on the OLED)
Pac-Man / Ghost	Game roles: prey (user-controlled) / autonomous hunters
PlatformIO	Build and flash tool-chain used inside the IDE
Scoreboard (Spielstandsanzeige)	External ESP8266 LED-matrix display that shows the game countdown timer etc.

2 System Requirements and Installation

The MAUSI software is split into three firmwares that are built and flashed with PlatformIO: the robot firmware running on the RP2040, the controller-bridge firmware running on the ESP32, and the scoreboard firmware running on the ESP8266. A shared library provides the common communication codes used by them.

2.1 Development Environment

- Operating system: Windows or macOS.
- IDE: CLion 2025.2 with the PlatformIO plugin. Visual Studio Code with the PlatformIO extension can be used as well. Required tool-chains are downloaded automatically by PlatformIO on the first build.
- Projects:
  - software/RP2040_Software: robot firmware. Built with the environment pico_normal.
  - software/xbox-esp32: ESP32 controller firmware. Default environment xiao_esp32s3 (Seeed XIAO ESP32-S3).
  - software/shared: shared library (communication codes), referenced by both projects via lib_extra_dirs.
  - software/spielstandsanzeige: scoreboard / game-timer firmware. Environment nodemcuv2 (NodeMCU v2, ESP8266). It drives a 32×8 LED matrix and shows the game countdown.

2.2 Flashing the Micromouse (RP2040)

Select the RP2040_Software folder in the IDE and build the project. There are two ways to flash the result onto the robot:

Option A, USB drag-and-drop (UF2):

- Put the robot into bootloader (BOOTSEL) mode. There are two ways to do this:
  - on the assembled robot, either press the reset button twice quickly or,
  - hold button B, briefly press reset, and then release B.
- The robot now appears as a USB mass-storage drive (RPI-RP2).
- Copy / drag the generated .uf2 file (.pio/build/pico_normal/firmware.uf2) onto that drive. The robot reboots automatically and runs the new firmware.

Option B, Direct upload from the IDE:

- Use the PlatformIO Upload action in CLion or VS Code. PlatformIO resets the board and flashes it over USB. If the board is not detected, a USB-to-serial driver may be required. Install it, or use Option A (UF2 drag-and-drop) instead.

2.3 Flashing the ESP32 Firmware

- Open the project software/xbox-esp32 and connect the ESP32 directly with a USB-C cable.
- Build and flash it via the PlatformIO Upload action. If the board is not detected, a USB-to-serial driver may be required.
- The ESP32 firmware uses the Bluepad32 Arduino core (configured through platform_packages) to connect to an Xbox controller over Bluetooth.
- The game ID is stored persistently on the RP2040 and pushed to the ESP32 at boot, so the ESP32 does not need to be re-configured after flashing.

2.4 Flashing the Scoreboard (ESP8266)

Open the project software/spielstandsanzeige, connect the ESP8266 via USB and flash it with the PlatformIO Upload action. The NodeMCU board needs a USB-to-serial driver.

2.5 Hardware Setup and Wiring (ESP32 <-> Micromouse)

The ESP32 is mounted on the robot and connected over UART to the expansion header that sits closer to the display. The ESP32 antenna connector is located next to Pin 6. Connect the four wires as listed below.

Signal	ESP32 pin	Micromouse header pin
UART data (ESP TX -> Mouse RX)	Pin 6 (TX)	Pin 28
UART data (ESP RX <- Mouse TX)	Pin 7 (RX)	Pin 29
Power 3V3	Pin 11 (3V3)	Pin 30
Ground	Pin 12 (GND)	Pin 31

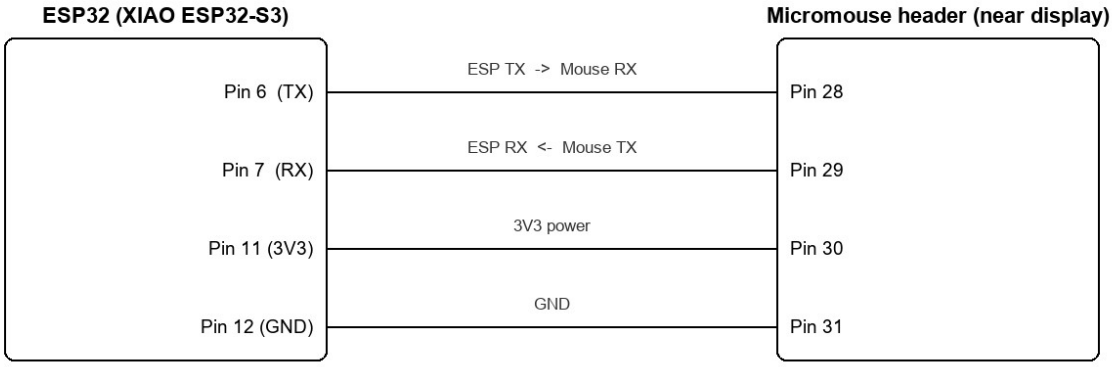


Diagram 1: UART connection between the ESP32 and the Micromouse expansion header.

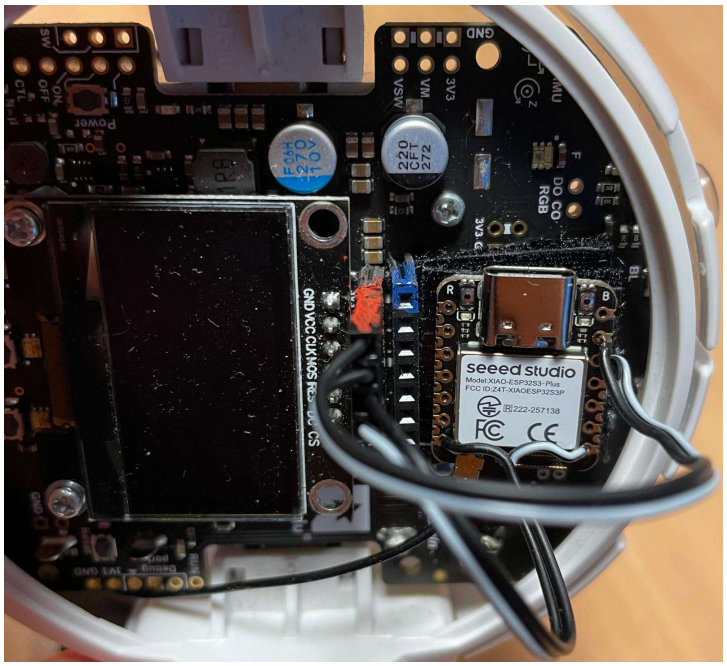


Figure 1: ESP32 (Seeed XIAO ESP32-S3) mounted on the robot and wired to the expansion header.

3 User Guide

After pressing the power button the robot shows the “MIKROMAUS-JAGD” splash logo for about two seconds and then enters the main menu. The whole robot is operated through the on-board 128x64 OLED display and the three buttons A, B and C. In game mode an Xbox controller (via the ESP32) is used in addition.

3.1 Controls

- Display: 128x64 OLED. The header line shows the current menu title, the selected mouse role (in the main menu), a blinking !AKKU warning on low battery, and a Bluetooth icon when a controller is connected. The bottom line shows the current button hints.
- Buttons in navigation mode: A = back, B = next (move down), C = OK / enter.
- Xbox controller (game mode only): the D-pad steers Pac-Man. The directions are absolute, as seen from above the maze (bird's-eye view), and not relative to the robot's own heading. Pressing up always moves Pac-Man toward the top of the maze. The X button activates Turbo. The ghosts drive fully autonomously and are not controlled manually.

3.2 Menu Structure

The following tree shows the complete on-screen menu. Sub-menus are marked with “>”, editable values and selectable options are shown on the right of each entry.

```
Main menu "Maus"   [header also shows: role name, [EDIT], !AKKU, BT icon]
|
+-- Karte                   map menu
|   +-- Kartierung          map the maze
|   +-- Anzeigen            show the stored map
|   +-- Senden              send map to other robots
|       +-- Wireless        via ESP32
|       +-- Serial          over USB / serial
|
+-- Loesen                  solve menu
|   +-- Ziel X              goal X        (0 .. MAZE_W-1)
|   +-- Ziel Y              goal Y        (0 .. MAZE_H-1)
|   +-- Starten             drive to the set goal
|   +-- Zufaellig           drive to a random goal
|
+-- Spiel                   game menu
|   +-- Demo                all robots autonomous
|   +-- Play Pacman         start the Pac-Man game
|
+-- Einstellung             settings menu
|   +-- Spieleinst.
|   |   +-- Modus           role: Pacman / Shadow / Speedy / Bashful / Pokey
|   |   +-- ID              robot ID      (0 .. 31)
|   |   +-- Start X         start node X  (0 .. MAZE_W-1)
|   |   +-- Start Y         start node Y  (0 .. MAZE_H-1)
|   +-- Turbo
|   |   +-- Zeit (s)        turbo duration
|   |   +-- Pause (s)       turbo cooldown
|   |   +-- Faktor %        turbo speed factor
|   +-- Solving
|   |   +-- Kurvenfahrt     Kurve (smooth curve) / Punkt (turn on the spot)
|   +-- Hardware
|   |   +-- ToF Kalibrieren calibrate the ToF sensors
|   |   +-- Kompass Ofs X   compass offset X
|   |   +-- Kompass Ofs Y   compass offset Y
|   |   +-- Reset Offsets   reset compass offsets
|   +-- Selbsttest
|   |   +-- Selbsttest      Aus / An
|   |   +-- Beeper          Aus / An
|   +-- BT Connect          connect the Bluetooth controller
|
+-- Test
|   +-- Selbsttest          run the self-test
|   +-- Batterie            battery voltage
|   +-- Akku-Lebensdauer    battery life test
|   +-- Kompass             compass test
|   +-- Linien              line sensors
|   +-- Gyroskop / Accel.   gyroscope / accelerometer
|   +-- Bumper              bumper test
|   +-- TOF                 ToF distance sensors
|   +-- Motoren             motor test
|   +-- LEDs                LED test
|
+-- Info                    firmware / system information
```

3.3 Operating the Robot

3.3.1 Map ("Karte")

- Kartierung: the robot drives through and learns the maze. The prompt “MAP START” asks you to place the robot on the start cross and keep it still while the compass heading is captured.
- Anzeigen: shows the currently stored map on the display.
- Senden: exports the stored map, either Wireless (via the ESP32) or Serial (over USB).

3.3.2 Solve ("Loesen")

- Set Ziel X and Ziel Y, then choose Starten to let the robot drive to that goal. Zufaellig picks a random goal.
- The Solving setting Kurvenfahrt selects how 90° turns are driven: Kurve = smooth curve through the intersection, Punkt = stop and turn on the spot.

3.3.3 Settings ("Einstellung")

- Spieleinst.: set the robot's Modus (role), its ID, and its Start X/Y node.
- Turbo: configure the Pac-Man X-button boost: Zeit (s) (duration), Pause (s) (cooldown) and Faktor % (speed factor).
- Hardware: calibrate the ToF sensors, set / reset the compass offsets.
- Selbsttest: enable the start-up self-test and its beeper. BT Connect connects the Bluetooth controller. Select it, then press the white Xbox button on the controller. The button blinks while pairing and turns solid white once connected. If it does not blink, the controller batteries are empty.

3.3.4 Test

- Runs individual hardware tests: self-test, battery, battery life, compass, line sensors, gyroscope, bumper, ToF, motors and LEDs. Info (main menu) shows firmware and system information.

3.4 Pac-Man Game Mode

3.4.1 Roles and Setup

- Three robots run at the same time: one Pac-Man and two ghosts: Shadow (red) and Speedy (pink). Each robot's role is selected under Einstellung > Spieleinst. > Modus.
- IDs: Pac-Man must have ID 0. Each ghost must have a non-zero ID, and the two ghosts must use different IDs.
- Every robot must already hold a maze map (run Kartierung first). A ghost refuses to start without a map.
- The start node of each robot is set via Start X / Start Y. All three robots must face North at the start.

3.4.2 Starting a Game

- On each ghost: either open Spiel and choose Play Pacman, or simply leave the ghost in the main menu. In both cases it waits for the start signal.
- On the Pac-Man robot: choose Spiel > Play Pacman. This opens a lobby screen that reads Xbox-Taste druecken zum Starten (press the Xbox button to start). The round begins only when the Xbox (Guide) button on the controller is pressed. Only then does Pac-Man broadcast the start signal to the ghosts, so all robots start at the same time and the ghosts do not move during setup. Pressing a mouse button (A, B or C) in the lobby aborts and returns to the menu.

3.4.3 Pac-Man HUD and Controls

- HUD: the remaining lives are shown as L:3 at the top. Six cheese icons along the left and right edges fill up as coins are collected (6 coins in total). A large central arrow points toward the nearest remaining coin, relative to the robot's current heading.
  DONE! appears once every coin is collected, and a shrinking bar at the bottom shows the remaining Turbo time.
- Controls: the D-pad sets the driving direction. The robot keeps driving in the latched direction until it reaches a wall or you choose a new direction. The X button activates Turbo, a temporary speed boost (configurable under Turbo).

3.4.4 Ghost Behaviour

- Shadow (red) chases Pac-Man's current position directly.
- Speedy (pink) predicts Pac-Man's position 2 to 4 nodes ahead along his current movement direction and aims for that point, trying to cut him off.
- The ghosts coordinate over the ESP32 link (cell reservations) so they do not collide with each other at intersections.

3.4.5 Capture, Lives and Victory

- A ghost captures Pac-Man when it is on the same node and its bumper is pressed. Pac-Man also loses a life if he drives into a ghost himself.
- On a capture, Pac-Man loses one life (of three). All robots then return to their start positions in a coordinated way and the round continues automatically.
- Victory (all coins collected): a victory melody plays as the last coin is collected and the screen shows VICTORY! with the coin count. All robots then return to their start positions and Pac-Man goes back to the lobby. Press the Xbox button to start a new round, or a mouse button (A, B or C) to return to the menu.
- If a ghost runs into a problem it shows an error screen (HINDERNIS! or SOLVER-FEHLER) with its position.

3.4.6 Scoreboard (Game Timer)

The scoreboard is a separate LED-matrix display. While idle it shows READY. When a Pac-Man game starts, it receives the start signal wirelessly and counts down from 2:30 minutes. While Pac-Man is being captured the countdown pauses and the display blinks, and it resumes when the next round starts. When it reaches 0:00, the game time is up, Pac-Man plays a short melody and every robot returns to its start position.

3.4.7 Troubleshooting

- If a robot freezes or the game logic gets stuck, press the reset button on the board and place that mouse back on its start node by hand.
- A ghost shows an error screen (HINDERNIS! or SOLVER-FEHLER) during the game: press button B on that ghost to dismiss the error and stop it. Place the ghost back on its start node by hand, facing North (always North, whatever the error was). Then open Spiel, choose Play Pacman and press C, so the ghost rejoins the running game while the other robots keep driving.
- A ghost loses its way while all robots are returning home: press button B on that ghost, then place it back on its start node facing North.
- Pac-Man loses its way while returning home: press button B and then button C on the Pac-Man robot.
- Pac-Man itself gets stuck during the game (rare corner case): reset all three robots and start a new game.
- A blinking red LED below the display means the battery is empty. Open the underside of the case and replace the batteries.

Appendix A: Spectator Game Flyer (Aushang)

Game flyer handed out to spectators at the Open Campus event.

MIKROMÄUSE JAGD

Schaffst du es, alle Käsestücke einzusammeln, bevor der Timer abgelaufen ist?

Eine echte **Robotermaus** wird zu **Pac-Man** und von dir per **Xbox-Controller** durchs Labyrinth gesteuert. Zwei weitere Mäuse sind die **Geister** und jagen dich automatisch. Sammle alle Käsestücke ein – aber lass dich nicht fangen!

1 Steuerung & Turbo

EINGABE	FUNKTION
Steuerkreuz	Robotermaus (Pac-Man) in die jeweilige Richtung fahren
X	Turbo aktivieren (Boost)
Xbox-Knopf	Spiel starten
Pfeil auf Robotermaus	Zeigt zum nächsten einzusammelnden Käsestück



Turbo-Mechanik

5 Sekunden Boost, danach **10 Sekunden Cooldown**, bevor der Turbo erneut genutzt werden kann.

5 s BOOST | 10 s COOLDOWN

2 Gewinnen, Gefangen & Spielstand

Gewinnen

Sammle **alle Käsestücke ein, bevor der Timer abgelaufen ist** – dann ist das Spiel gewonnen! ⏱ Anschließend fahren Pac-Man und beide Geister automatisch zurück an ihre Ausgangsposition.

Gefangen?

Erwischt dich ein Geist, starten **alle Mäuse wieder an ihren Ausgangspositionen** – das Spiel läuft sofort weiter.

Käse bleibt Käse

Beim Fangen geht **nichts verloren**: Dein bereits gesammelter Käse bleibt erhalten und der **Spielstand zählt weiter**, bis der Timer abgelaufen ist.

3 Die Jäger – zwei Geister

VERFOLGER
Roter Geist

Fährt Pac-Man **direkt hinterher** und versucht ihn auf dem kürzesten Weg zu fangen.

ABFÄNGER
Lila Geist

Fährt Pac-Man ein Stück **voraus**, um ihm den Weg abzuschneiden und ihn in die Zange zu nehmen.

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Viel Erfolg – schnapp dir den Käse!

Appendix B: Operator Quick Guide (Helfer-Anleitung)

Three-page operator guide used by the helpers at the stand.

MIKROMÄUSE JAGD · HELFER-ANLEITUNG

Spiel einrichten & starten – Schritt für Schritt

Diese Anleitung zeigt, wie das Pac-Man-Spiel mit den drei Mikromäusen aufgebaut und gestartet wird und was zu tun ist, wenn einmal etwas hakt. Einfach der Reihe nach abarbeiten.

1 Alle Mäuse einschalten

Drücke den **Power-Knopf** an **allen drei Mäusen** (Pac-Man + beide Geister). Die Maus startet und zeigt auf dem Display das **Startmenü**.

Akku-Hinweis

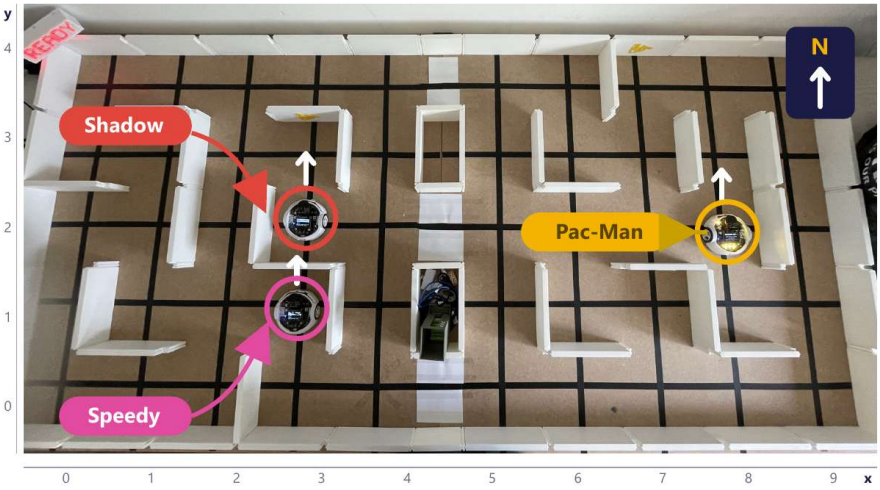
Blinkt nach dem Einschalten die **rote LED unterhalb des Displays**, ist der Akku leer – siehe Schritt 5.



Power-Knopf (linke Seite)

2 Geister auf die Startpositionen stellen

Stelle jede Maus auf ihren Startknoten. **Pac-Man rechts**, die beiden Geister links übereinander. Alle Mäuse müssen **nach Norden schauen** (zur oberen Spielfeldkante – siehe weiße Pfeile).



MAUS	FARBE	STARTKNOTEN (X, Y)
Pac-Man	gelb	**(8, 2)**
Shadow	rot	**(3, 2)**
Speedy	pink	**(3, 1)**

↑ Blickrichtung Norden

Alle drei Mäuse **nach Norden** ausrichten (zur oberen Spielfeldkante). Der Knoten **(0, 0)** liegt **links unten**; x läuft nach rechts, y nach oben.

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3 Pac-Man mit dem Controller verbinden

1. Nach dem Einschalten zeigt die **Pac-Man-Maus** das **Startmenü**. (hier aus Bild nicht korrekt)
2. Mit Taster **B** nach unten navigieren bis **„Spieleinstellungen"**, mit Taster **C** bestätigen (OK).
3. Mit Taster **B** weiter bis **„Bluetooth Connect"**, mit Taster **C** bestätigen. Die Maus geht in den **Kopplungsmodus**.
4. Am **Xbox-Controller** die **weiße Xbox-Taste** drücken.
5. Das Xbox-Symbol **blinkt** (Kopplung läuft) und leuchtet danach **dauerhaft weiß** = verbunden.
6. Mit Taster **A** (Zurück) zurück ins **Hauptmenü**.
7. Im Hauptmenü auf **„Spiel"** → **„Play Pac-Man"**.
8. Zum Loslegen den **Xbox-Knopf** am Controller drücken → das **Spiel startet**.

⚠ Symbol blinkt nicht?

Blinkt das Xbox-Symbol beim Drücken nicht, sind vermutlich die **Batterien des Controllers leer** –> tauschen und erneut versuchen.

🎮 Menü steuern (auf Pac-Man)

Wahlweise mit den **Tastern auf der Maus** (B = weiter, C = OK, A = zurück) **oder** mit dem **Xbox-Controller**: mit dem **Steuerkreuz** nach oben/unten, **Taste B** = bestätigen, **Taste A** = zurück.



Taster auf der Platine

A zurück **B** weiter ▼ **C** OK



Weiße Xbox-Taste – zum Koppeln & Starten

4 Spielablauf

▶ Läuft von allein

Das Spiel läuft nach dem Start **automatisch bis zum Ende** durch. Danach kann mit dem **Xbox-Knopf** direkt ein **neues Spiel** gestartet werden.

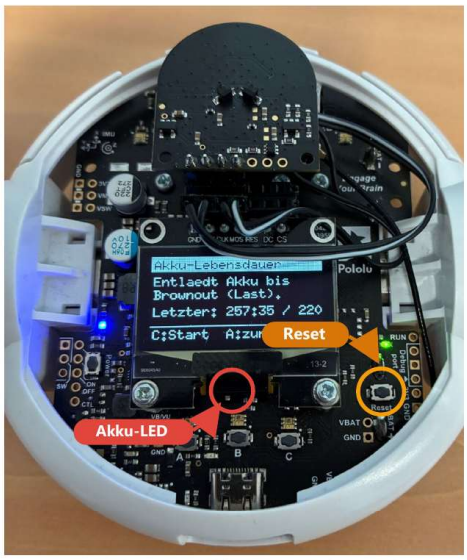
5 Wenn etwas hakt

Eine Maus wirft einen Fehler / die Spiellogik stockt:
Den **Reset-Taster** (rechts neben dem Display) drücken, die Maus wieder auf ihren **Startknoten** setzen -> sie läuft weiter.

Der Fehler bleibt:
Alle Mäuse zurücksetzen und ein **neues Spiel** beginnen.

🔋 Akku leer

Blinkt die **rote LED unterhalb des Displays**, ist der Akku leer: Gehäuse an der **Unterseite** öffnen und **Batterien tauschen**.



Reset (rechts) & rote Akku-LED (unter dem Display)